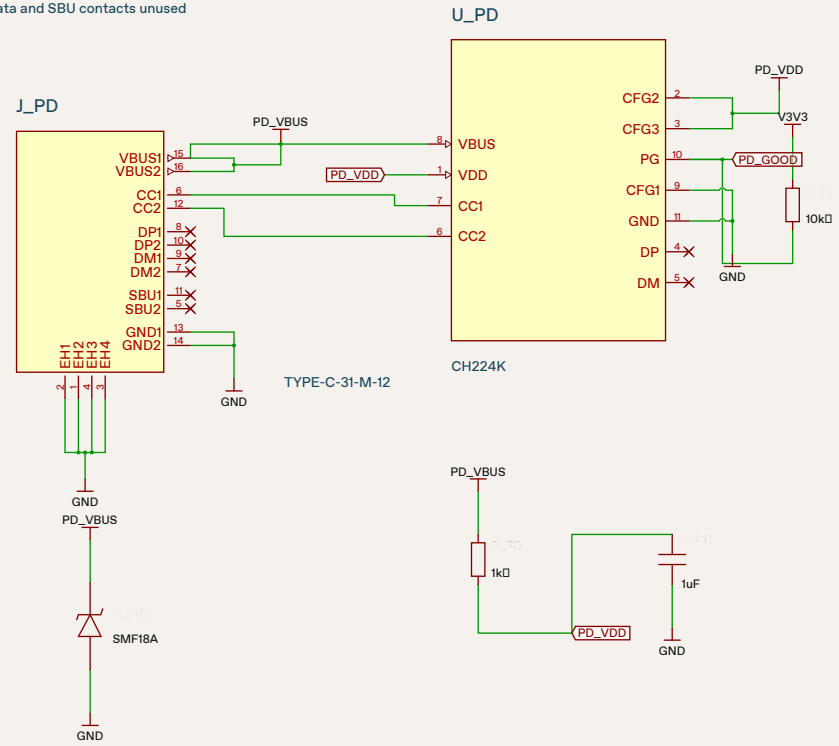
NEMA14 / RP2040

Power & USB-PD

Motor supply negotiation, reverse-blocked logic power and 3.3 V conversion

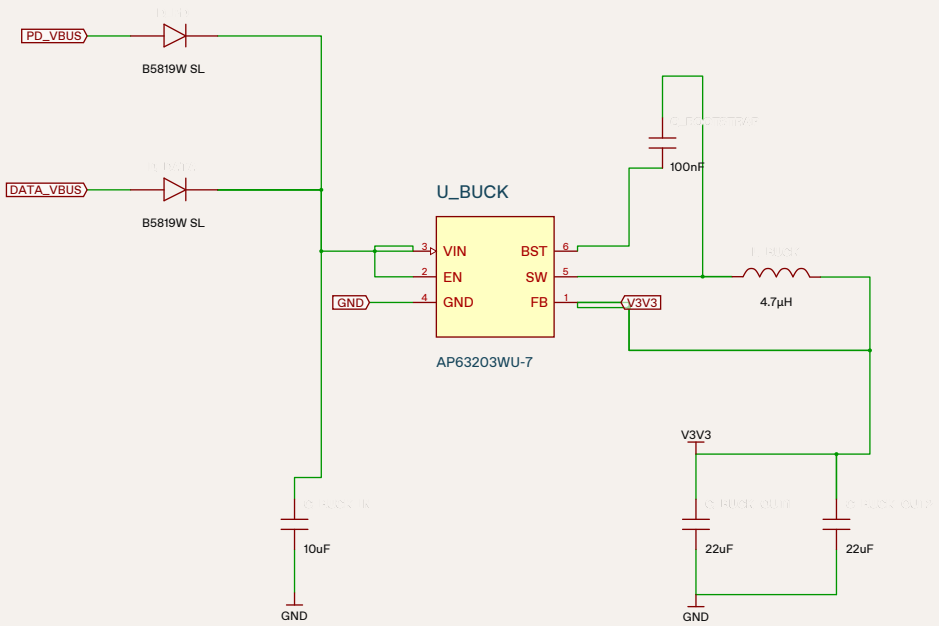
USB-C motor power

Fixed 15 V request; data and SBU contacts unused



3.3 V logic supply

AP63203 fixed-output buck; USB power is diode-isolated



PD_VBUS = motor rail DATA_VBUS = programming USB 5 V

VIN range and transients require bench qualification.

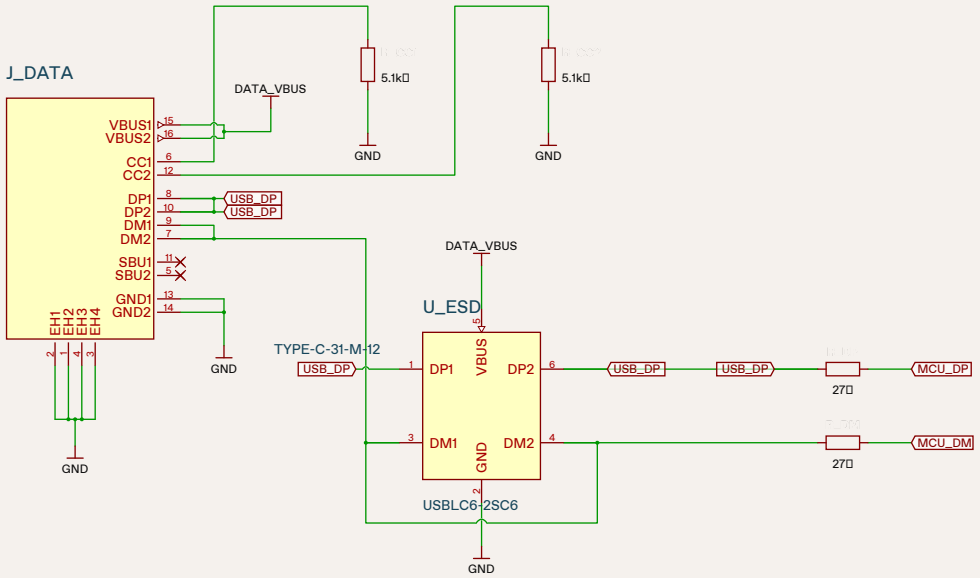
NEMA14 / RP2040

Programming & USB

Programming USB-C, data-line protection and nonvolatile boot interfaces

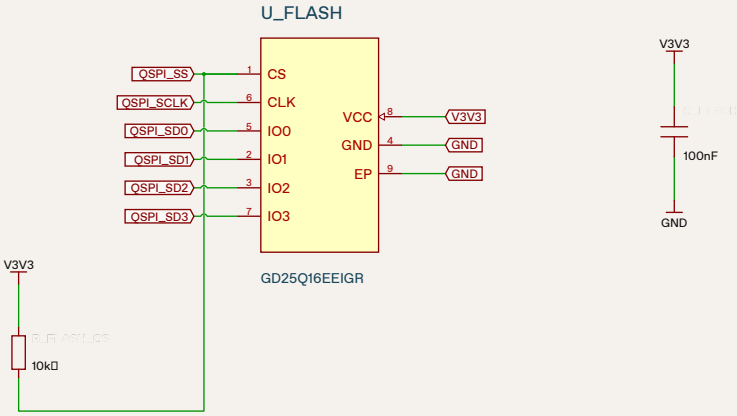
USB-C data port

Separate 5.1 kΩ CC pull-downs; duplicated D+/D− contacts



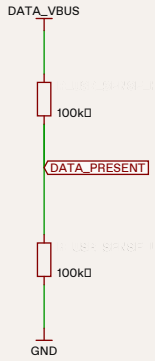
QSPI flash & boot selection

GD25Q16E; BOOT contact asserts the boot input



USB presence

Resistive divider to MCU; fixed 5 V input only





AnasSarkiz / RP2040 motor controller

35 × 35 mm | 4-layer PCB | Hardware prototype — assembly/firmware/bench qualification pending

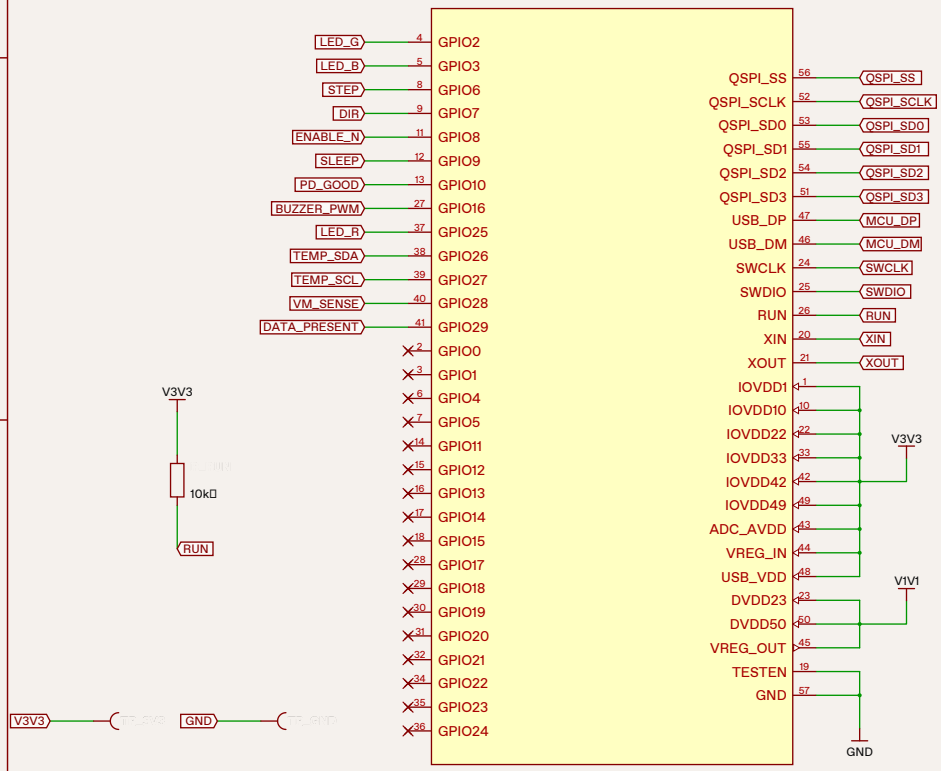
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Controller & clocks

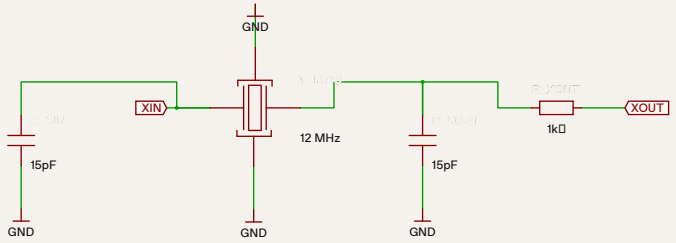
MCU interfaces, reset/debug access, power decoupling and clock support

U_MCU



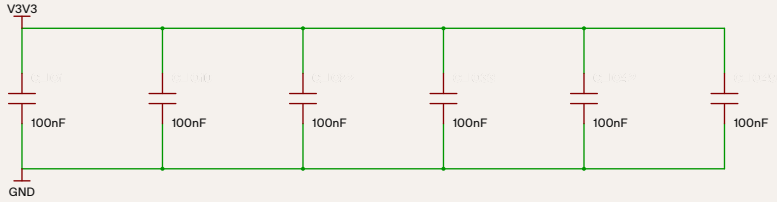
12 MHz crystal

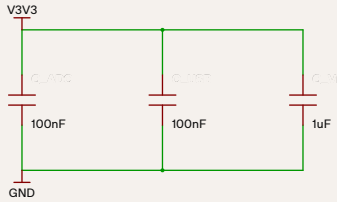
Native 4-pin crystal; load capacitors and series resistor



3.3 V bypass

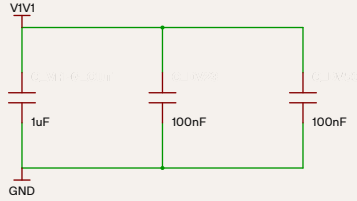
IOVDD, ADC and USB decoupling





1.1 V core

On-chip VREG_OUT feeds DVDD



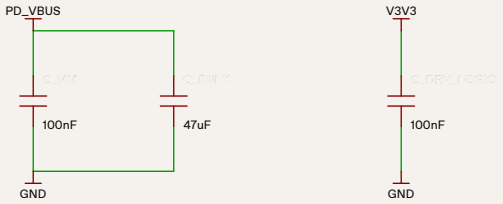
NEMA14 / RP2040

Stepper motor driver

A4988 phase outputs, charge pump, current limit and motor-voltage measurement

Driver supplies

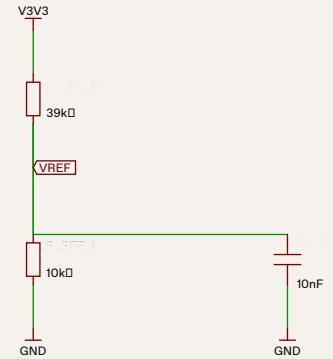
Local bulk and high-frequency bypass

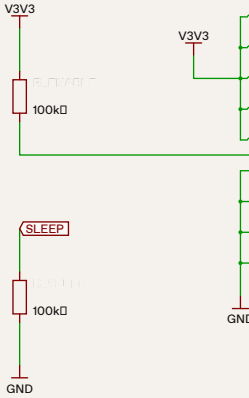


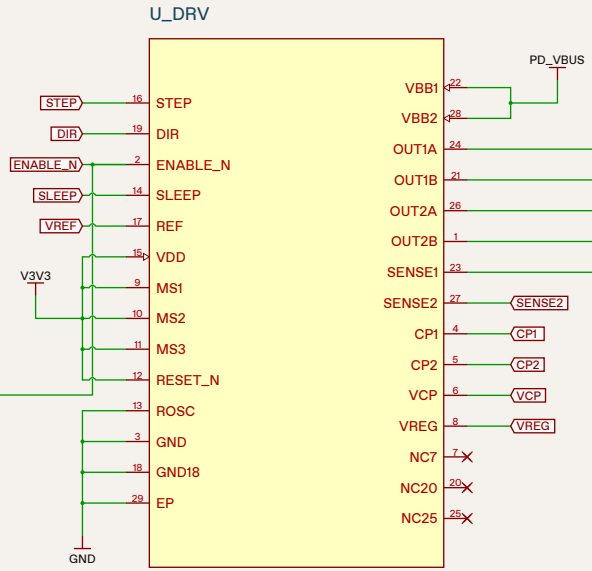
JST PH pin order: 1 A+ 2 A– 3 B+ 4 B–

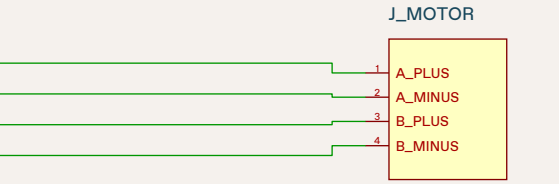
Current reference

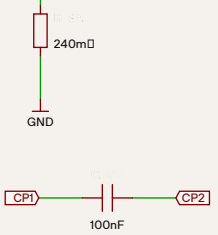
VREF / (8 × RSENSE) = 0.351 A nominal

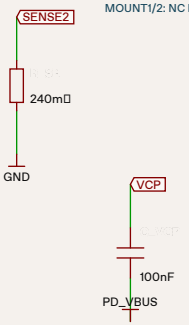






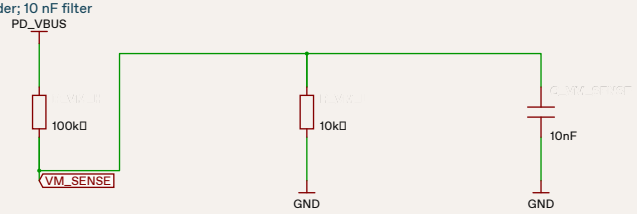






Motor voltage monitor

100 kΩ / 10 kΩ divider; 10 nF filter





AnasSarkiz / RP2040 motor controller

35 × 35 mm | 4-layer PCB | Hardware prototype — assembly/firmware/bench qualification pending

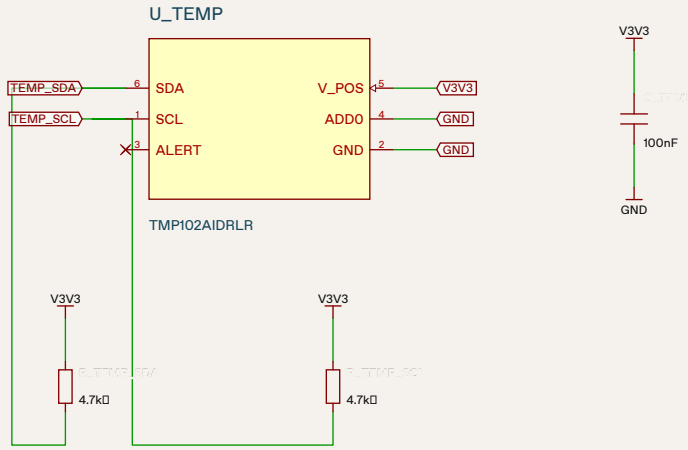
NEMA14 / RP2040

Temperature & feedback

I²C temperature telemetry, RGB status outputs and PWM magnetic buzzer

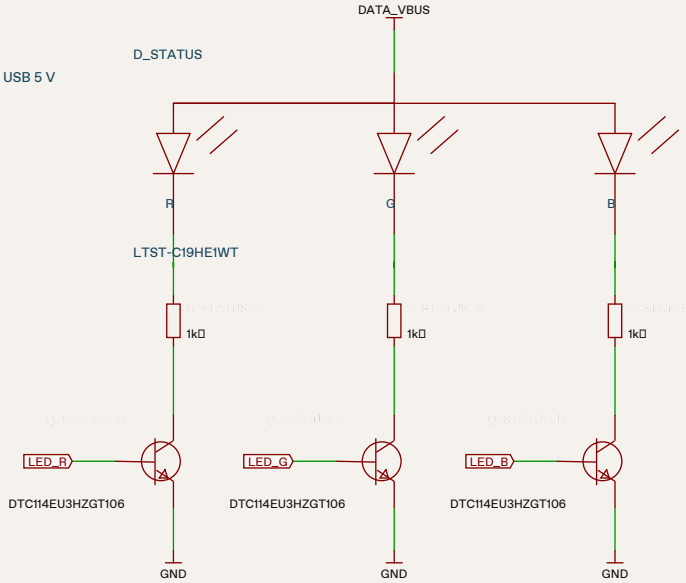
TMP102 temperature

ADD0 grounded: address 0x48; ALERT unused



RGB status indicator

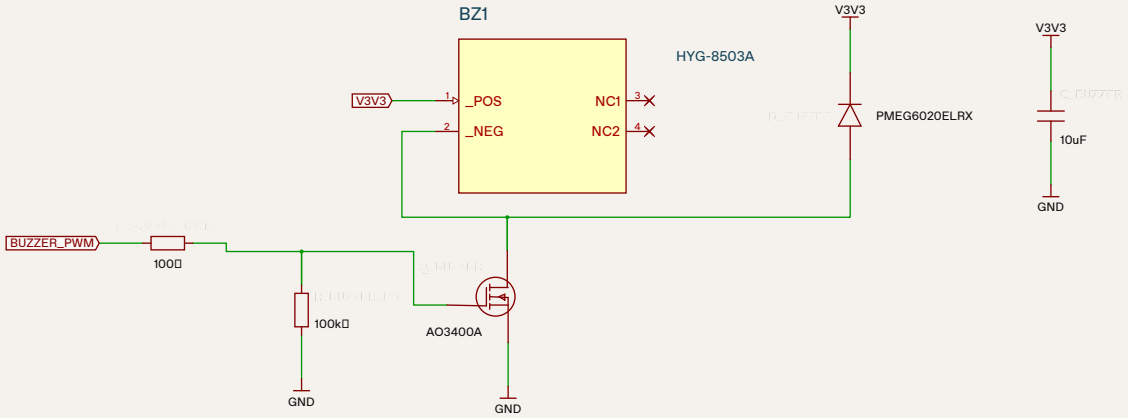
Common anode uses programming USB 5 V



DTC114E: NPN with internal input/bias resistors.

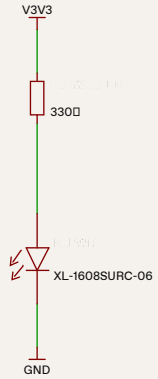
PWM buzzer

120 mA magnetic load; MOSFET drive and flyback diode



Power indication

3.3 V rail present; RGB/buzzer require firmware



AnasSarkiz / RP2040 motor controller

35 × 35 mm | 4-layer PCB | Hardware prototype — assembly/firmware/bench qualification pending

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